

module4_lab2

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1 Module 4 Lab 2: Textbook Problem 3.1

1.1 The Bookstore Inventory

This lab uses the bookstore inventory data to find the median cover type, the mean weight of all books, and the mean weight for each cover type.

GitHub repository: <https://github.com/roee0910/cs82a-portfolio>

The data file contains 15 books. The **weight** column is measured in grams, and **cover** identifies the book as hardback (hb) or paperback (pb).

```
[1]: import pandas as pd

books = pd.read_csv("allbacks.csv")
books.head()
```

```
[1]:
```

	rownames	volume	area	weight	cover
0	1	885	382	800	hb
1	2	1016	468	950	hb
2	3	1125	387	1050	hb
3	4	239	371	350	hb
4	5	701	371	750	hb

1.2 1. Median cover type

Because the cover types are ordered categories (hb comes before pb), I can sort them and take the middle value. There are 15 books, so the median is the 8th value after sorting.

```
[2]: cover_sorted = books["cover"].sort_values().reset_index(drop=True)

print("Sorted cover types:")
print(cover_sorted.to_string(index=False))
print("\nNumber of books:", len(cover_sorted))
print("Middle position:", (len(cover_sorted) + 1) // 2)
print("Median cover type:", cover_sorted.iloc[(len(cover_sorted) - 1) // 2])
```

Sorted cover types:

hb
hb

hb
hb
hb
hb
hb
pb
pb
pb
pb
pb
pb
pb
pb

Number of books: 15
Middle position: 8
Median cover type: pb

The median cover type is **pb (paperback)** because the 8th cover type in the ordered list is pb.

```
[3]: median_cover = cover_sorted.iloc[(len(cover_sorted) - 1) // 2]
      median_cover
```

```
[3]: 'pb'
```

1.3 2. Mean weight of all books

The overall mean weight is the total weight of all 15 books divided by 15.

```
[4]: overall_mean_weight = books["weight"].mean()

      print("Total weight:", books["weight"].sum(), "grams")
      print("Number of books:", books["weight"].count())
      print("Mean weight:", overall_mean_weight, "grams")
```

Total weight: 10600 grams
Number of books: 15
Mean weight: 706.6666666666666 grams

The mean weight of all the books is **706.67 grams**. This means a typical book in this inventory weighs about 707 grams on average.

```
[5]: overall_mean_weight
```

```
[5]: np.float64(706.6666666666666)
```

1.4 3. Mean weight by cover type

Next, I grouped the books by cover type and calculated the mean weight for each group.

```
[6]: cover_means = books.groupby("cover")["weight"].mean()

print(cover_means.round(2))
```

```
cover
hb    796.43
pb    628.12
Name: weight, dtype: float64
```

The mean weight is **796.43 grams for hardback books** and **628.13 grams for paperback books**. Hardback books are about **168.30 grams heavier on average** than paperback books, which suggests that the type of cover is associated with a noticeable difference in book weight in this inventory.

```
[7]: difference = cover_means["hb"] - cover_means["pb"]

print("Hardback mean:", round(cover_means["hb"], 2), "grams")
print("Paperback mean:", round(cover_means["pb"], 2), "grams")
print("Difference:", round(difference, 2), "grams")
```

```
Hardback mean: 796.43 grams
Paperback mean: 628.12 grams
Difference: 168.3 grams
```

1.5 4. By-hand verification

I verified the overall mean using the formula:

Mean = total of all weights ÷ number of books

The total weight is 10,600 grams and there are 15 books, so:

10,600 ÷ 15 = 706.67 grams

This matches the Python result of 706.67 grams.

For the median, there are 15 ordered cover values, so the middle position is the 8th value. The first 7 are hb and the 8th is pb, so the median cover type is pb, which also matches the Python result.

1.6 Final answers

- **Median cover type: pb** (paperback)
- **Mean weight of all books: 706.67 grams**
- **Mean weight of hardback books: 796.43 grams**
- **Mean weight of paperback books: 628.13 grams**
- **Difference between group means: 168.30 grams**, with hardback books being heavier on average.

The calculations and the by-hand checks agree, and the group comparison suggests that hardback books in this inventory tend to weigh more than paperback books.